

Kevin Benny Mathew

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EDUCATION

2023 - 2027	B.E. (Hons.) Electronics and Communication Engineering - BITS Pilani, Goa Campus	8.21/10 (Upto Semester 4)
May 2023	ISC (Class XII) - The Bishop's School, Camp	97.6%
May 2021	ICSE (Class X) - St. Mary's School, Pune	99.0%

SKILLS & PROFICIENCY

Languages	Verilog, C, C++, Python, Java
Key Courses	VLSI Design, Digital Design, Microprocessors & Interfacing, Digital Signal Processing, Control Systems, Communication Systems, Operating Systems (Ongoing), Computer Architecture (Ongoing)
Technologies	Microcontrollers (ESP32, Teensy, ESP8266, Arduino), ROS/ROS2, Linux, Git
Tools	KiCAD, Vivado, Cadence Virtuoso, PlatformIO, MATLAB, Fusion 360, Solidworks, Onshape, Android Studio

WORK EXPERIENCE

Summer Intern 	May 2025 - Aug 2025
- Worked at the Indian Meteorological Department on hardware-software integration and algorithm development. - Developed embedded firmware for ESP8266 hardware using C++ and PlatformIO. - Implemented custom fuzzy logic and Kalman filters for high-accuracy sensor data acquisition. - Conducted data analysis using Python and documented results for research publication.	

PROJECTS

Project Kratos: Mars Rover (Electronics Subsystem) 	Aug 2024 - Present
- Redesigned the electronics hardware architecture for modularity, easier assembly, and improved debugging. - Migrated the control stack from ROS Noetic to ROS2 Humble. - Designed custom mounts for cameras and sensors using Fusion 360.	
Low Cost Electromechanical Tensile Tester	May 2025 - Dec 2025
- Executed end-to-end development including electronics architecture, mechanical design, and 3D modeling. - Integrated motion control systems using Teensy microcontrollers, stepper motors, and a custom UI. - Developed a horizontal tensile tester for the characterization of soft materials under Dr. Tushar Sakorikar.	
Project Vulcan: Humanoid Robot 	Feb 2024 - Present
- Implemented audio amplification circuits for microphones located in the robot's ears. - Led the bionic arm subsystem, developing a 7-DOF biomimetic robotic arm emulating human movement. - Improved internal mechanical systems for facial movements, specifically eyes and mouth.	

Development of Flexible Tactile Sensors

Dec 2025 - Present

– Developing tactile sensors for robotic perception under Dr. Tushar Sakorikar.

– Project is currently in literature review and ideation phase with plans to start fabrication by February 2026.

LEADERSHIP & VOLUNTEERING

Research Head - Electronics and Robotics Club (ERC), BITS Goa

Apr 2025 - Present

- Guiding **75+** students across multiple robotics projects (Vulcan, Snakebot, Quadruped)

- Managing project progress and integrating electronics, mechanical, and automation verticals.

- Organized and designed ROS2-based hackathon with **200+ participants nationwide**

Committee Member - Sandbox Innovation Lab, BITS Goa

Feb 2024 - Present

Maintaining lab equipment of the campus student-run (3D printers, laser cutters) and managing tech club purchases.

Volunteer - Abhigyaan, BITS Goa

Oct 2023 - Present

Provided evening tuition classes for K-12 children of campus non-teaching staff.

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